

Power BI Data Preparation and Ranking Model

Step-by-Step Methodology Guide for a Future Analyst

A2: YouTube Intro to Ranking Systems

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Purpose of This Document

This document serves as a complete methodological guide and internal manual for the YouTube Trending Video Ranking System built in Power BI. It is written for a future analyst who may need to understand, audit, refine, or extend the ranking algorithm and the underlying data pipeline without reverse engineering in the PBIX file.

The document covers the full analytical workflow: from data ingestion and cleaning (ETL), through the statistical reasoning behind normalization and feature construction, to the final composite ranking formula and dashboard design. Every major design decision is explained with its rationale, so that the pipeline remains transparent, reproducible, and intentional.

Section 1: ETL Process and Data Preparation

This section documents every transformation applied to the raw data inside Power Query Editor, the rationale behind each step, and the downstream implications for the ranking model.

Step 1: Data Ingestion

Two datasets were imported into Power BI from the provided Dropbox sources:

- **CAvideos.csv** - the main fact table containing one row for each video-trending-date combination (approximately 40,000 rows). Each row records engagement metrics (views, likes, dislikes, comment count) and metadata (title, channel, publish time, tags) for a video on the specific date it appeared in the Canadian YouTube Trending section.
- **CA_category_id.json** - a reference lookup table mapping each numeric category_id to its human-readable category name (e.g., 25 = News & Politics, 10 = Music). This file also contains the is_assignable field, which indicates whether YouTube creators can manually assign their videos to that category.

Step 2: Structural Error Removal

During initial inspection of the CSV file, a structural parsing issue was identified: certain rows contained embedded line breaks within quoted fields, which caused Power BI to misalign columns.

The issue was resolved in Power Query by navigating the source step and enabling the option to ignore quoted line breaks. This ensured that multi-line text values were treated as single fields rather than triggering row splits.

Step 3: Header Promotion

The first row of the CSV was promoted to column headers. This is a standard step when importing flat files where Power BI does not automatically detect the header row. After promotion, the dataset correctly showed field names.

Step 4: Date Column Preparation

The trending_date column appeared in an abbreviated format (e.g., "17.14.11" for November 14, 2017) that was incompatible with Power BI's native Date type. The following transformation sequence was applied:

- The column was split.
- The two-digit year values were prefixed with "20" to produce four-digit years (e.g., "17" became "2017").
- The three components were merged back into a single column in YYYY-MM-DD format.

- The resulting column was converted to Date type.

Step 5: Data Type Verification and Correction

Rather than relying on Power BI's automatic type detection, all columns were manually verified and corrected. This is critical because auto-detection can silently misclassify fields, leading to broken relationships or incorrect aggregations downstream.

Column	Assigned Type	Rationale
video_id	Text	Identifier, not a measurable quantity
trending_date	Date	Reconstructed via split-merge (Step 4)
title	Text	Descriptive metadata
channel_title	Text	Descriptive metadata
category_id	Text	Must match JSON lookup type for relationship
publish_time	DateTime	ISO 8601, natively recognized
tags	Text	Pipe-separated keywords, kept for future use
views	Whole Number	Quantitative engagement metric
likes	Whole Number	Quantitative engagement metric
dislikes	Whole Number	Quantitative engagement metric
comment_count	Whole Number	Quantitative engagement metric
comments_disabled	True/False	Boolean flag for dynamic weight logic
ratings_disabled	True/False	Boolean flag for dynamic weight logic
video_error_or_removed	True/False	Boolean flag for exclusion filter

Step 6: Column Removal

Columns that did not contribute to the ranking algorithm were removed.

Step 7: Category Cleaning - Filters

Category_id 29 was found to be missing from the lookup table, yet some videos in the CSV referenced it. These orphan videos appeared under (Blank) when filtered by category name and displayed zero values for all engagement metrics, distorting category-level aggregations.

To address these issues, the following page-level filters were applied:

- category_name is not (Blank) - excludes videos with no matching category
- category_id is not 29 - explicitly removes the category
- video_error_or_removed = False - removes inaccessible videos

Step 8: Relationship Construction

A relationship was created between the two tables:

- **From:** CAvideos[category_id] (many side)
- **To:** CA_category_id[category_id] (one side)
- **Cardinality:** Many-to-one (*:1)
- **Cross-filter direction:** Single (from lookup to fact table)

Single direction was chosen over bidirectional because bidirectional filtering introduces ambiguity in DAX evaluation contexts and can produce unexpected results when multiple slicers interact. Single direction provides predictable, auditable filter propagation.

Step 9: Duplicate Handling Strategy

The dataset contains what initially appears to be duplicate rows: the same video_id appears multiple times. However, these are not true duplicates. Each row represents a distinct trending date for that video. A video that trended for five consecutive days will have five rows, with different engagement metrics on each date (as views, likes, and comments accumulate over time).

These rows were intentionally preserved because:

- They contain the temporal dimension necessary for calculating Days Trending, a key component of the ranking score.
- They allow the dashboard to show the correct engagement snapshot for each specific trending date.
- A DaysTrending_Lookup table was created to track how many days each video has been trending as of any given date.

Step 10: Date Table Construction

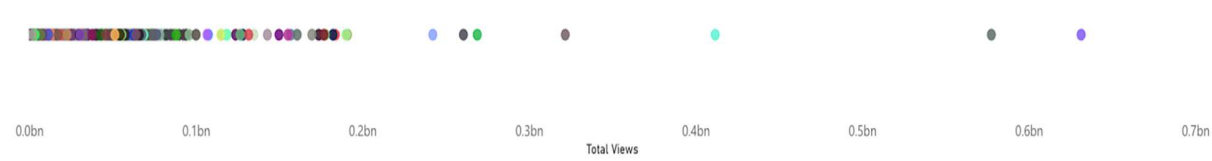
A dedicated DateTable was created to support time-based filtering and analysis. The table contains calculated columns: Day, MonthName, Year, Day, Week of day and IsWeekend. This is a standard dimensional modeling practice that decouples date logic from the fact table, enabling the Trending Date slicer to filter by year, month, or weekday without additional transformations on the raw data. The DateTable was marked as the official date table in Power BI to enable time intelligence functions if needed in future iterations.

Section 2: Statistical Methodology and Normalization

This section explains the statistical reasoning behind the normalization techniques evaluated and the final method chosen. The goal is to ensure that no single metric dominates the composite ranking score purely due to its numerical scale, while preserving meaningful differences between videos.

Main metrics distribution: (see the dominance of outliers in all the metrics.)

Extreme variation in views shows that raw values would dominate the ranking score without logarithmic scaling



Dislikes are concentrated near zero with a few strong reactions indicating engagement outliers



Likes show strong right-skewness confirming the presence of high-magnitude engagement outliers



Comments are unevenly distributed across videos, highlighting differences in active engagement levels



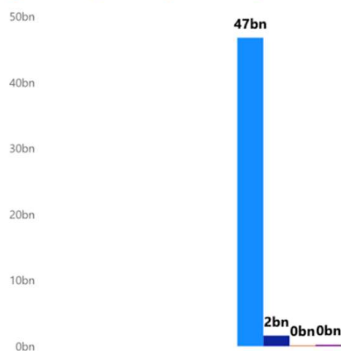
2.1 The Scale Problem – Why normalize?

Raw engagement variables operate on very different numerical scales, requiring normalization before combining them into a ranking score

2bn Total Likes	80M Total Dislikes	203M Total Comments	46.776373446bn Total Views
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Raw engagement variables operate on drastically different magnitudes, which would bias the ranking score without normalization

● Total Views ● Total Likes ● Total Dislikes ● Total Comments



Without normalization the ranking score would be dominated by views due to their much larger magnitude. Normalization ensures that each engagement variable contributes comparably to the final ranking score.

The four engagement variables operate on fundamentally different numerical scales:

If these raw values were combined directly into a weighted sum, views would dominate the ranking score regardless of the weights assigned, simply because its magnitude is orders of magnitude larger. A video with 10 million views and zero comments would rank higher than a video with 500,000 views and 50,000 comments, defeating the purpose of a multi-dimensional ranking system.

This observation motivated the need for normalization: a mathematical transformation that brings all variables onto a comparable scale while preserving their relative ordering.

2.2 Normalization Methods Evaluated

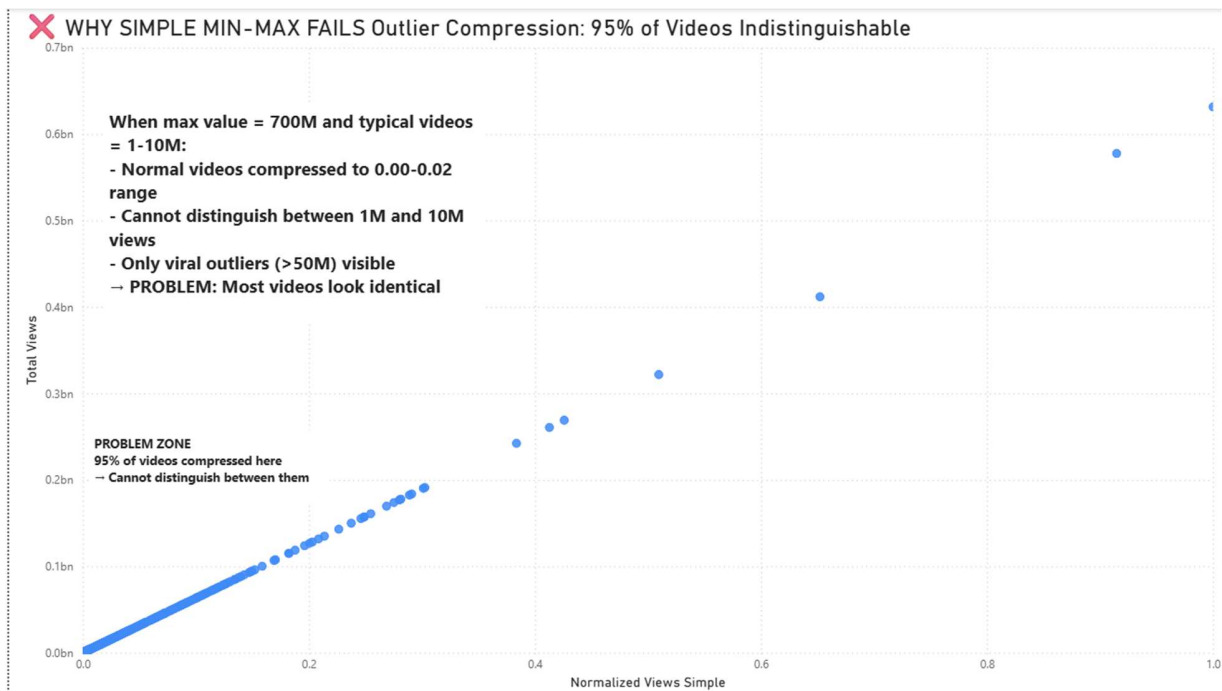
Four normalization approaches were tested and compared. The PBIX file contains dedicated pages that visualize the distribution of engagement variables under each method.

Simple Min-Max Normalization

Formula: $(X - \text{Min}) / (\text{Max} - \text{Min})$, producing values in $[0, 1]$.

This method is straightforward but suffers from extreme sensitivity to outliers. In a dataset where the maximum views value is 200 million and the median is 500,000, virtually all videos are compressed into the 0.00-0.05 range, with only a handful of viral outliers near 1.0. The resulting normalized scores have almost no discriminative power for the vast majority of videos.

Verdict: Rejected. Fails to differentiate between videos in the middle of the distribution, which is where the ranking needs the most granularity.

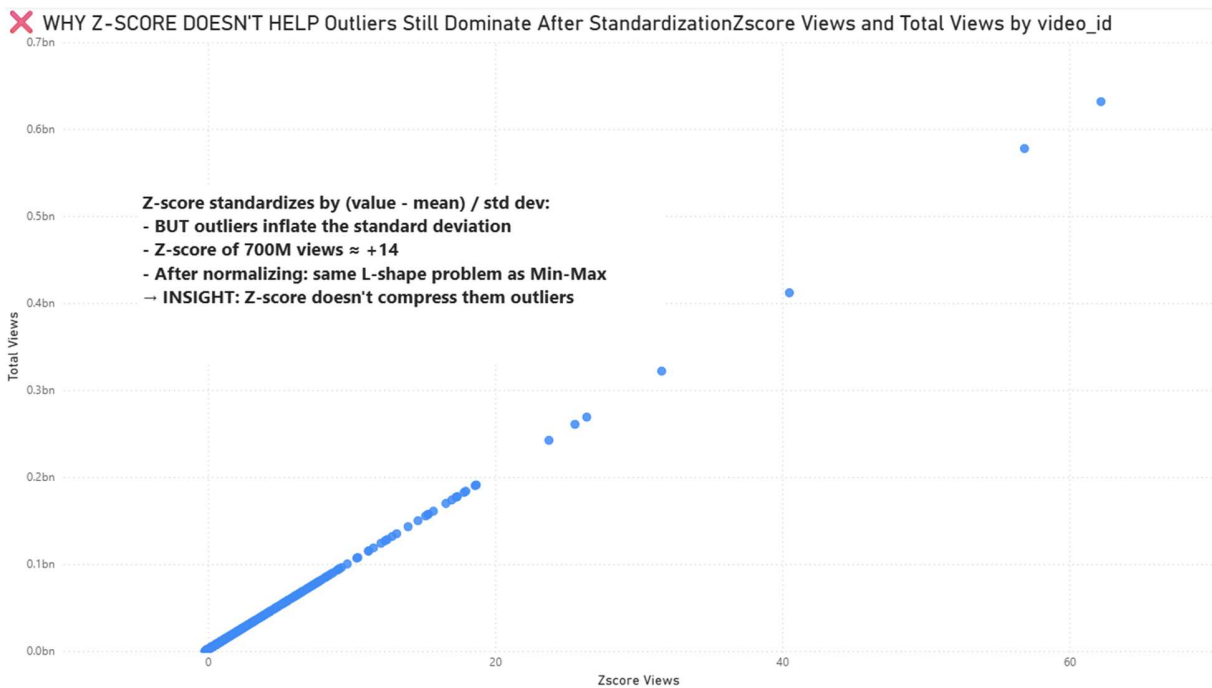


Z-Score Normalization

Formula: $(X - \text{Mean}) / \text{Standard Deviation}$, producing values centered around 0.

Z-scores are effective when data follows an approximately normal distribution. However, YouTube engagement metrics are heavily right skewed with extreme outliers. Z-scores produce negative values for most videos (since the mean is pulled upward by outliers), and the resulting values are unbounded, making it difficult to compare contributions from different metrics.

Verdict: Rejected. Unbounded output and poor behavior with skewed distributions make Z-scores unsuitable.



EX. DAX FOR Z-SCORE

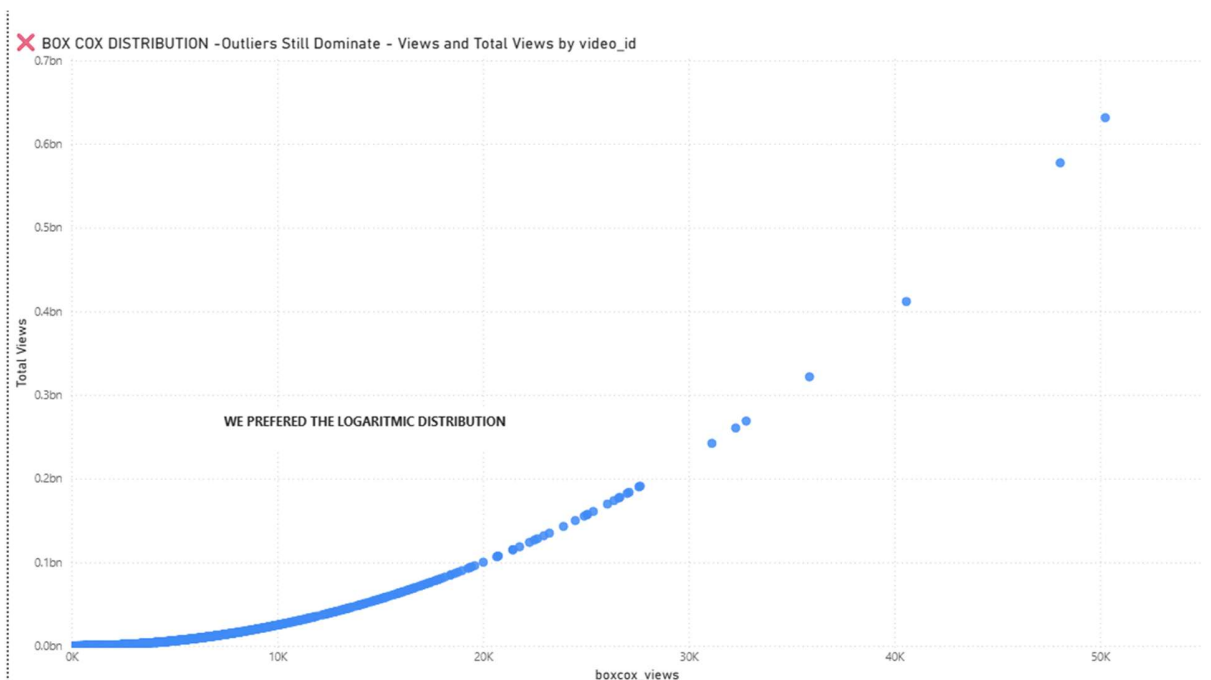
Zscore Views =

```
VAR bc_table = ADDCOLUMNS(ALL(CAvideos[video_id]), "val", [Total Views])
VAR mean = AVERAGEX(bc_table, [val])
VAR std = STDEVX.P(bc_table, [val])
RETURN DIVIDE([Total Views] - mean, std)
```

Box-Cox Transformation

Box-Cox applies a power transformation parameterized by lambda to approximate normality. While theoretically powerful, it requires parameter estimation, cannot handle zero values without an additive constant, and does not natively produce bounded output. The additional complexity does not provide sufficient benefit for this application.

Verdict: Rejected. Overly complex for the marginal improvement; parameter estimation adds fragility.



EX. DAX FOR BOX COX

boxcox_likes =

VAR y = [Total Likes]

VAR lambda = 0.5

RETURN

IF(lambda = 0, LN(y), (POWER(y, lambda) - 1) / lambda)

Normalized BoxCox Likes =

VAR bc_table = ADDCOLUMNS(ALL(CAvideos[video_id]), "bc_val", [boxcox_likes])

VAR bc_min = MINX(bc_table, [bc_val])

VAR bc_max = MAXX(bc_table, [bc_val])

RETURN DIVIDE([boxcox_likes] - bc_min, bc_max - bc_min)

Logarithmic (LOG10) Normalization - Selected Method

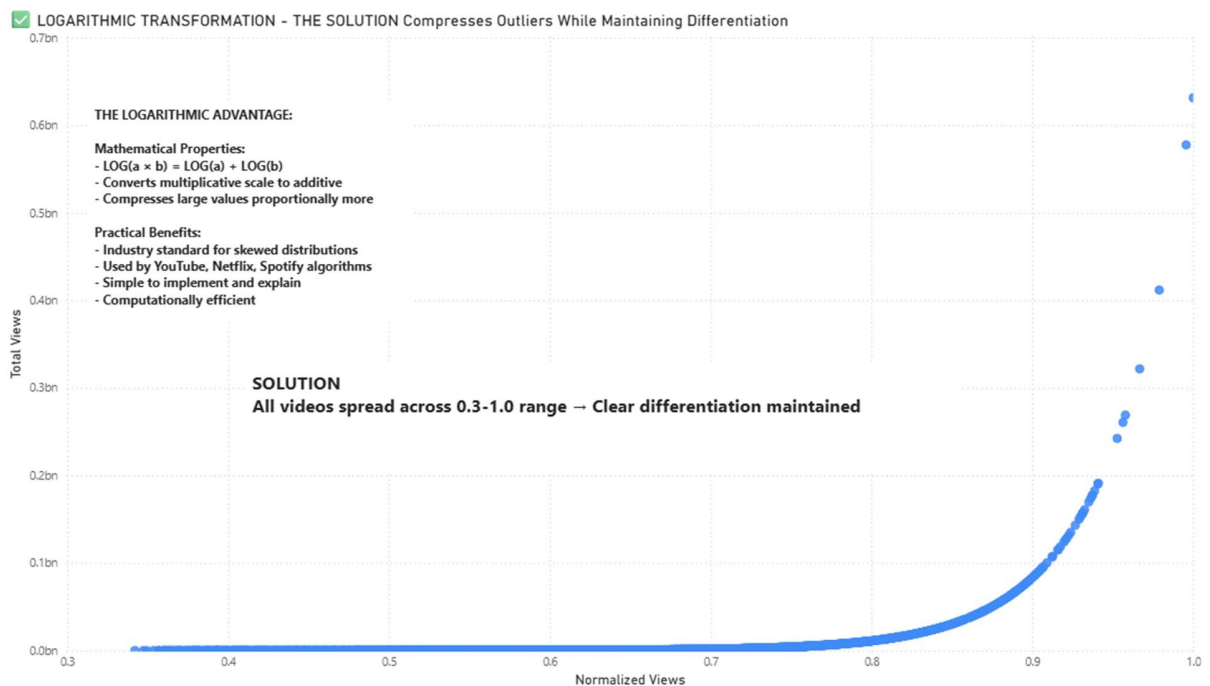
Formula: $\text{LOG10}(X + 1) / \text{MAX}(\text{LOG10}(X + 1))$, producing values in [0, 1].

The logarithmic transformation compresses extreme values at the top of the distribution while expanding the lower and middle ranges, directly addressing the right-skew problem. The +1 constant ensures that zero values produce a valid result ($\text{LOG10}(1) = 0$). Division by the observed maximum rescales the output to [0, 1].

This method was selected because:

- It handles the extreme right-skew of engagement data naturally.
- It produces bounded output in [0, 1], enabling straightforward weighted combination.
- It preserves ordinal ranking; more views still score higher.
- It prevents any single outlier from compressing the rest of the distribution.

- It is transparent, simple to audit, and requires no parameter estimation.



Section 3: Ranking Algorithm Design

This section documents the complete ranking formula, the rationale for each weight, and the dynamic handling of missing data.

3.1 Base DAX Measures

The following base measures aggregate the raw engagement data:

```
Total Views = SUM(CAVideos[views])
Total Likes = SUM(CAVideos[likes])
Total Dislikes = SUM(CAVideos[dislikes])
Total Comments = SUM(CAVideos[comment_count])
```

Days Trending is derived from the DaysTrending_Lookup table, which counts distinct trending dates for each video up to the selected date.

3.2 Normalized Measures (LOG10)

Each base measure is transformed using the LOG10 normalization formula:

```
Normalized Views =
IF(
    ISEMPTY(CAVideos),
    BLANK(),
    VAR LogViews = LOG10([Total Views] + 1)
    RETURN
    DIVIDE(LogViews, [Max Log Views], 0))
```

Where [Max Log Views] calculates the maximum LOG10 value across the entire filtered dataset:

```

Max Log Views =
CALCULATE(
    MAXX(
        VALUES(CAVideos[video_id]),
        VAR VideoViews = [Total Views]
        RETURN LOG10(VideoViews + 1)
    ),
    ALL(CAVideos)
)

```

The same pattern is applied for Normalized Likes, Normalized Dislikes, and Normalized Comments. All normalized values are within the range [0, 1].

3.3 Engagement Composite Sub-Score

The engagement metrics are grouped into a composite sub-score with intentional weight differentiation:

Base weights: Comments 50% | Dislikes 30% | Likes 20%

Comments (50%): Commenting represents the highest level of user engagement. It requires active cognitive effort: reading the content, forming an opinion, and typing a response. High comment activity signals genuine audience interaction.

Dislikes (30%): A dislike is an **active engagement signal**. Unlike a passive view, a dislike requires conscious evaluation. Videos with high dislikes are often controversial but highly engaging. In a ranking system designed to surface engaging content, dislikes are a meaningful signal, not a penalty.

Likes (20%): While likes indicate positive reception, they require the least cognitive effort among active engagement signals. A like is a single-click action that may reflect casual approval rather than deep engagement.

3.4 Dynamic Weight Redistribution for Disabled Features

Some videos have `comments_disabled = True` and/or `ratings_disabled = True`. For these videos, the corresponding metrics are 0, but this zero does not reflect low engagement - it reflects impossibility. Treating these zeros at face value would unfairly penalize the video.

The Normalized Engagement measure dynamically redistributes weights based on available data:

```

Normalized Engagement =
VAR HasComments = NOT(SELECTEDVALUE(CAVideos[comments_disabled], TRUE))
VAR HasRatings = NOT(SELECTEDVALUE(CAVideos[ratings_disabled], TRUE))

```

```

-- Available weight pools
VAR CommentWeight = IF(HasComments, 0.50, 0)
VAR DislikeWeight = IF(HasRatings, 0.30, 0)
VAR LikeWeight = IF(HasRatings, 0.20, 0)
VAR TotalWeight = CommentWeight + DislikeWeight + LikeWeight

```

```

RETURN
IF(
    TotalWeight = 0,
    BLANK(), -- No engagement data available at all
    ([Normalized Comments] * CommentWeight +
    [Normalized Dislikes] * DislikeWeight +
    [Normalized Likes] * LikeWeight) / TotalWeight
)

```

```

Ranking Score LOG =
VAR EngValue = [Normalized Engagement]
VAR HasEngagement = NOT(ISBLANK(EngValue))

VAR ViewWeight = 0.10
VAR EngWeight = IF(HasEngagement, 0.80, 0)
VAR TrendWeight = 0.10
VAR TotalWeight = ViewWeight + EngWeight + TrendWeight

RETURN
((([Normalized Views] * ViewWeight) +
(IF(HasEngagement, EngValue, 0) * EngWeight) +
([Normalized Days Trending] * TrendWeight)) / TotalWeight

```

This mechanism works as follows:

- All available :base weights apply (50/30/20).
- Comments disabled: Dislikes becomes 60%, Likes becomes 40% (rescaled from 30/20 out of 50).
- Ratings disabled: Comments receives 100%.
- Both disabled: Engagement returns BLANK; Ranking Score adjusts accordingly.

3.5 Final Ranking Score

```

Ranking Score LOG =
VAR EngValue = [Normalized Engagement]
VAR HasEngagement = NOT(ISBLANK(EngValue))

VAR ViewWeight = 0.10
VAR EngWeight = IF(HasEngagement, 0.80, 0)
VAR TrendWeight = 0.10
VAR TotalWeight = ViewWeight + EngWeight + TrendWeight

RETURN
((([Normalized Views] * ViewWeight) +
(IF(HasEngagement, EngValue, 0) * EngWeight) +
([Normalized Days Trending] * TrendWeight)) / TotalWeight

```

Views (10%): deliberately low weight. High view counts can be driven by clickbait, autoplay, or algorithmic promotion, none of which reflect content quality. If views dominated, the algorithm would replicate a sort-by-views.

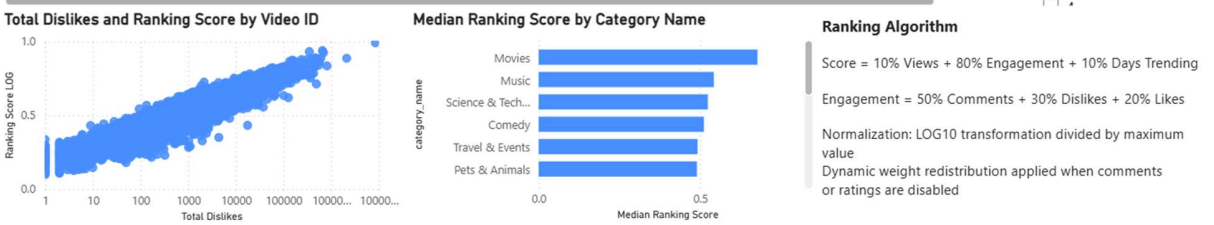
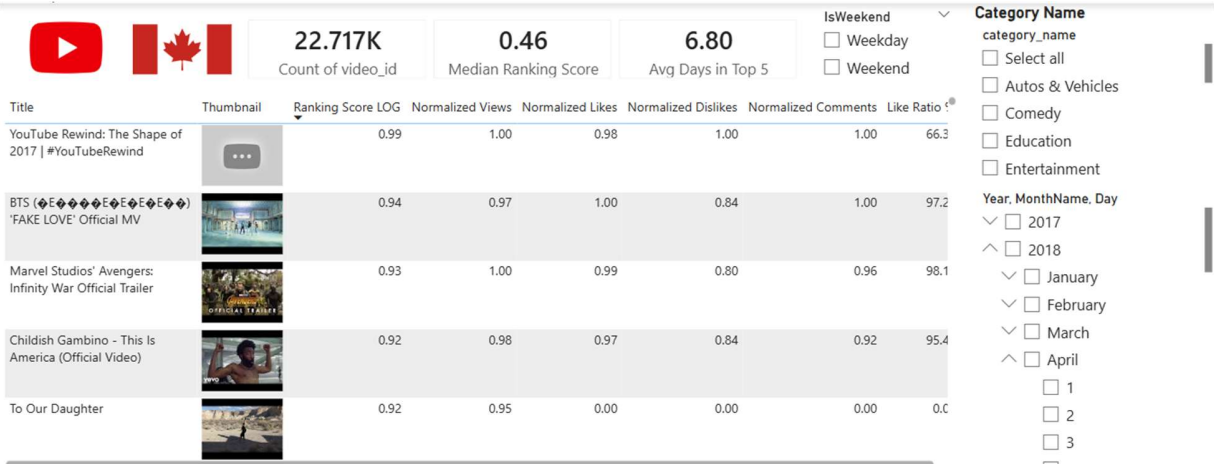
Engagement (80%): dominant weight. Comments, likes, and dislikes all require conscious user action, making them stronger signals of content relevance than passive views.

Days Trending (10%): modest reward for persistence. A video trending for multiple days has demonstrated sustained interest, but the weight is low to avoid over-rewarding older videos.

When Engagement is BLANK (both features disabled), the same redistribution applies: Views and Days Trending are rescaled to 50/50.

Section 4: Dashboard Design

2 Dashboards



Statistical Methods Comparison: Ranking Score Analysis

category_name

☐ Select all

☐ Autos & Vehicles

☐ Comedy

☐ Education

☐ Entertainment

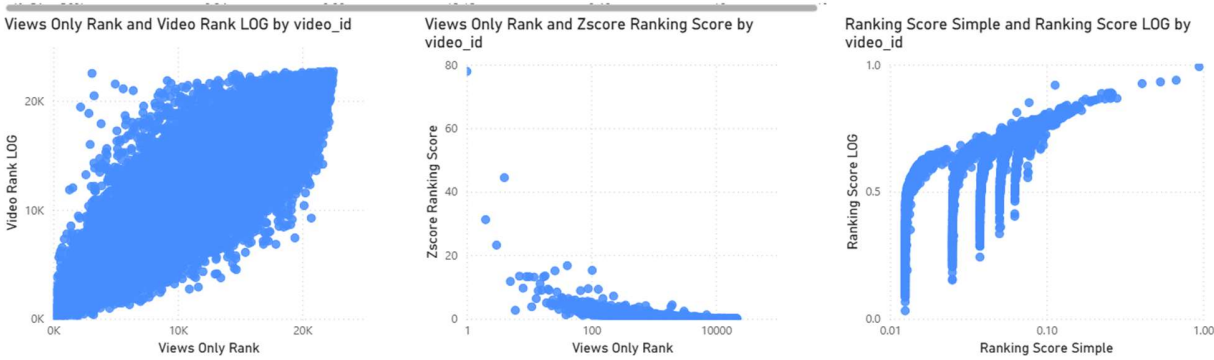
☐ Film & Animation

☐ Gaming

Year, MonthName, Day

☒ 2017

☐ 2018



4.1 Dashboard Overview

This dashboard presents a YouTube trending video ranking system built for the Canadian market, covering the period from November 2017 to June 2018. The system ranks trending videos based on a composite engagement score designed to reflect multiple dimensions of video performance beyond simple view counts.

The ranking algorithm combines three components: Normalized Views (10%), Normalized Engagement (80%), and Normalized Days Trending (10%). The Engagement metric is further broken down into Comments (50%), Dislikes (30%), and Likes (20%), reflecting the principle that active audience participation is a stronger quality signal than passive consumption. All metrics are transformed using LOG10 normalization to reduce the influence of extreme outliers before being combined into a final ranking score.

The dashboard consists of two main pages. The first page displays the Leaderboard, showing the top five recommended videos for any selected trending date and category. Users can filter the results by category, date, and day type (weekday or weekend) using the interactive slicers. Each video entry includes its title, thumbnail, ranking score, normalized engagement metrics, and like ratio. The second page provides a statistical comparison across four different ranking methodologies, allowing users to evaluate how the choice of normalization method affects the final ranking output.

The system also handles edge cases dynamically. When a video has comments or ratings disabled, the algorithm automatically redistributes the weights across the remaining available metrics, ensuring that all videos receive a fair and comparable ranking score regardless of their data availability.

4.2 Analysis

This analysis of the YouTube Canada trending dataset produced six key insights through interactive exploration of the dashboard.

Insight 1: Algorithm vs. Views-Only Ranking

Although a positive correlation is observed between the algorithm-based ranking and the views-only ranking, a considerable number of data points deviate from the diagonal line. This suggests that the engagement-weighted algorithm produces rankings that differ meaningfully from a simple views-based ordering, reflecting a broader set of performance dimensions beyond reach alone.

Insight 2: Justification for LOG Normalization over Z-score

The distribution of Z-scores revealed that a small number of videos produced extremely high values, indicating strong sensitivity to outliers. This observation prompted the choice of LOG normalization as the principal transformation method, as it efficiently diminishes the impact of extreme values while maintaining relative disparities among videos. The scatter plot comparing Simple and LOG scores further shows an S-curve distribution, confirming that LOG transformation redistributes scores more evenly across the dataset.

Insight 3: Dynamic Weight Redistribution in Practice

Within the main dashboard, the video "To Our Daughter" ranks third despite showing values of 0.00 for Likes, Dislikes, and Comments. This reflects a case where ratings and comments were disabled, and the dynamic weight redistribution mechanism responded by reallocating the weights to the remaining available metrics, demonstrating the robustness of the algorithm under incomplete data conditions.

Insight 4: Weekend vs. Weekday Content Preferences

Weekend trending videos show a notably different category of distribution compared to weekdays. Movies and Pets & Animals rank more prominently during weekends, suggesting that viewers tend to consume longer-form or leisure-oriented content when they have more free time. In contrast, weekdays show stronger performance in Science & Technology and How-to & Style, indicating a preference for informational and educational content during working days. This pattern suggests that content creators and platform recommender systems could benefit from adjusting their recommendation strategies based on the day of the week.

References

Milli, S., Carroll, M., Wang, Y., Pandey, S., Zhao, S., & Dragan, A. D. (2023). Engagement, user satisfaction, and the amplification of divisive content on social media. Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT). <https://knightrcolumbia.org/content/engagement-user-satisfaction-and-the-amplification-of-divisive-content-on-social-media>